
Online Examination Report

Date:	28.07.2014
Attempts:	1
Examinee:	Mbuya 2, Denis
Date of birth:	11/10/1986
Subject:	080- Principles of flight
Result:	0 %
Time used:	00:00:18 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	195	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
2	231	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
3	192	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
4	199	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
5	214	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
6	207	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
7	184	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
8	205	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
9	148	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
10	188	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
11	210	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
12	187	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
13	229	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
14	215	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
15	225	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
16	190	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
17	212	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
18	82	81.1.1.1	Laws and definitions	0,00	1,00
19	37	81.1.1.3	Aerodynamic forces and moments on aerofoils	0,00	1,00
20	4	81.1.4.2	Induced drag	0,00	1,00
21	57	81.1.4.2	Induced drag	0,00	1,00
22	28	81.1.8.5	Special phenomena of stall	0,00	1,00
23	26	81.1.10.1	Spoilers and the reasons for use	0,00	1,00
24	9	81.2.2.2	Oblique shock waves	0,00	1,00
25	14	81.4.1.1	Basics and definitions	0,00	1,00
26	41	81.4.1.1	Basics and definitions	0,00	1,00
27	75	81.4.3.14	Factors affecting the man stab/stick force per g	0,00	1,00
28	15	81.4.4.4	Factors affecting static directional stability	0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	5	81.4.5.5	Factors affecting static lateral stability	0,00	1,00
30	63	81.4.6.3	Dutch roll	0,00	1,00
31	72	81.5.4.5	Means to avoid adverse yaw:	0,00	1,00
32	10	81.5.8.3	Stabiliser trim	0,00	1,00
33	106	81.7	PROPELLERS	0,00	1,00
34	50	81.7.1.3	Fixed and variable pitch/constant speed	0,00	1,00
35	73	81.7.1.4	Propeller efficiency versus speed	0,00	1,00
36	65	81.8.1.2	Straight steady climb	0,00	1,00
37	69	81.8.1.2	Straight steady climb	0,00	1,00
38	24	81.8.1.2	Straight steady climb	0,00	1,00
39	29	81.8.1.4	Straight steady glide	0,00	1,00
40	92	81.8.3	Particular points on the polar curve	0,00	1,00
Total:0%				0,00	40,00

Attachment 2: List of result by topic

Licence

Date: 28.07.2014

Examinee: Mbuya 2, Denis

Location: Uganda Civil Aviation Authority

Your Result: 0,00 from 40,00 Points (0,00 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
81. PRINCIPLES OF FLIGHT AEROPLANE	40	0,00	40,00	0,00
81.1. SUBSONIC AERODYNAMICS	6	0,00	6,00	0,00
81.2. HIGH SPEED AERODYNAMICS	1	0,00	1,00	0,00
81.4. STABILITY	6	0,00	6,00	0,00
81.5. CONTROL	2	0,00	2,00	0,00
81.7. PROPELLERS	3	0,00	3,00	0,00
81.8. FLIGHT MECHANICS	5	0,00	5,00	0,00
Total	40	0,00	40,00	0,00

1

When an aircraft's forward CG limit is exceeded, it will affect the flight characteristics of the aircraft by producing

- ☒ **higher stalling speeds and more longitudinal stability.**
- ☐ **improved performance since it reduces the induced drag.**
- ☐ **very light elevator control forces which make it easy to inadvertently overstress the aircraft.**

2

If both the ram air input and drain hole of the pitot system are blocked, what airspeed indication can be expected?

- ☐ Decrease of indicated airspeed during a climb.
- ☒ No variation of indicated airspeed in level flight even if large power changes are made.
- ☐ Constant indicated airspeed during a descent.

3

An aircraft is loaded with the CG aft of the aft limit. What effect will this have on controllability?

- ☒ Stall and spin recovery may be difficult or impossible.
- ☐ A stall will occur at a higher indicated airspeed due to the greater downloading on the elevator.
- ☐ A stall will occur at a lower airspeed, but recovery will be easier because of reduced wing loading.

4

The best lift/drag ratio of a glider is a value that

- ☐ varies depending upon the weight being carried.
- ☒ remains constant and is independent of the weight being carried.
- ☐ remains constant regardless of airspeed changes.

5

What is the suggested speed to fly when passing through lift with no intention to work the lift?

- ☐ Best glide speed.
- ☐ Best lift/drag speed.
- ☒ Minimum sink speed.

6

At what point during an autotow should the glider pilot establish the maximum pitch attitude for the climb?

- ☒ 200 feet above the ground.
- ☐ Between 300 and 400 feet above the ground.
- ☐ 100 feet above the ground.

7

If the nosewheel of an airplane moves aft during gear retraction, how would this aft movement affect the CG location of that airplane? It would

- ☐ have no effect on the CG location.
- ☒ cause the CG location to move aft.
- ☐ cause the CG location to move forward.

8

What could result if a glider pilot releases while in the low-tow position during an aerotow?

- ☒ Tow ring may strike and damage the glider after release.
- ☐ Nose of the glider would tend to pitch up after release.
- ☐ Glider may be forced into the towplane's wake turbulence.

9

When a glider is turning in flight, the force that opposes the inward turning force is called

- ☐ adverse yaw.
- ☒ centrifugal force.
- ☐ resultant force.

10

In a twin-engine airplane, the single-engine service ceiling is the maximum density altitude at which V(YSE) will produce

- ☒ 50 feet per minute rate of climb.
- ☐ 500 feet per minute rate of climb.
- ☐ 100 feet per minute rate of climb.

11

A rule of thumb for flying a final approach is to maintain a speed that is

- ☐ twice the glider's stall speed plus half the estimated windspeed.
- ☐ 50 percent above the glider's stall speed, regardless of windspeed.
- ☒ 50 percent above the glider's stall speed plus half the estimated windspeed.

12

Which characteristic of a spin is not a characteristic of a steep spiral?

- ☐ Rapid loss of altitude.
- ☐ High rate of rotation.
- ☒ Stalled wing.

13

How does the wing design feature 'washout' affect the production of lift?

- ☐ The center of lift moves from the trailing edge of the wing, to the leading edge of the wing, as the wing begins to stall.
- ☐ The main body of the wing continues to produce lift when the wing tips are not producing lift.
- ☒ The wing tips continue producing lift when the main body of the wing is not producing lift.

14

Which would most likely ensure a safe off-field landing?

- ☐ Landing in a pasture or uncultivated field rather than one in cultivation and whose crops have been harvested.
- ☐ Landing into the wind, regardless of the type or slope of the terrain.
- ☒ Maintaining an approach airspeed of at least 50 percent above the glider's stall speed plus half the estimated windspeed.

15

As a weight shift aircraft wing approaches a stall, the wing tips

- ☐ act in much the same way as ailerons on a three-axis aircraft.
- ☒ decrease the wings angle of attack.
- ☐ increase the wings angle of attack.

16

Maximum gliding distance of an aircraft is obtained when

- ☒ induced drag and parasite drag are equal.
- ☐ induced drag equals the coefficient of lift.
- ☐ parasite drag is the least.

17

With regard to two or more gliders flying in the same thermal, which statement is true?

- ☐ All turns should be to the right.
- ☒ Turns should be made in the same direction as the first glider to enter the thermal.
- ☐ Turns should be in the same direction as the highest glider.

18

The SI units of air density (I) and force (II) are..... &... respectively:

☐ (I) N / kg, (II) kg.

☒ (I) kg / m³, (II) N.

☐ (I) N / m³, (II) N.

☐ (I) kg / m², (II) kg.

19

The angle between the direction of the undisturbed airflow (relative wind) and the chord line of an aerofoil is called:

☐ same as the angle between chord line and fuselage axis.

☒ angle of attack.

☐ climb path angle.

☐ glide path angle.

20

Which statement concerning the local flow pattern around a wing is correct?

- ☐ Sweepback reduces drag since, because the span increases.
- ☐ Slat extension, at a constant angle of attack and normal extension speeds, will increase the lift coefficient,
- ☒ By fitting winglets to the wing tip, reduces induced drag.
- ☐ Vortex generators on the wing partially block the spanwise flow over the wing

21

The induced drag:

- ☐ increases as the magnitude of the tip vortices decreases.
- ☒ increases as the lift coefficient increases.
- ☐ has no relation to the lift coefficient.
- ☐ increases as the aspect ratio increases.

22

A stick pusher:

- ☐ pushes the elevator control forward prior to stick shaker activation.
- ☐ pushes the elevator control to avoid a stall at a negative load factor.
- ☒ pushes the elevator control forward when a specified value of angle of attack is exceeded.
- ☐ vibrates the elevator control.

23

When a wing spoiler is extended at constant angle of attack:

- ☒ drag increases but lift decreases.
- ☐ drag increases but lift remains constant.
- ☐ both drag and lift decrease.
- ☐ both drag and lift increase.

24

Which of these statements about an oblique shock wave are correct or incorrect?

- I. The local speed of sound in front of an oblique shock wave is higher than behind it.
II. The Mach number in front of an oblique shock wave is higher than behind it.

- ☒ I is incorrect, II is correct.
- ☐ I is incorrect, II is incorrect.
- ☐ I is correct, II is correct.
- ☐ I is correct, II is incorrect.

25

A statically unstable aeroplane:

- ☐ will become stable at high speed.
- ☐ is always dynamically stable.
- ☒ can never be dynamically stable.
- ☐ reduces its instability as the CG moves aft.

26

(Please view annex 081-0023)

Assuming no pilot input the motion of the aeroplane in the diagram shows:

- ☒ neutral dynamic longitudinal stability.
- ☐ dynamic longitudinal stability.
- ☐ static longitudinal instability.
- ☐ dynamic longitudinal instability.

See Attachments:

1. 081-0023.gif

Annex

(Attachment
No.1)

27

Stick force per g:

- ☒ is dependent on CG location.
- ☐ does not change with increasing altitude.
- ☐ has a maximum value related to acceptable controllability, the minimum value is of no concern.
- ☐ is selected by the pilot by electronic means before each flight.

28

Which of the following provides a positive contribution to static directional stability?

- ☒ A dorsal fin.
- ☐ Moving the centre of gravity aft.
- ☐ A low wing as compared with a high wing.
- ☐ A forward swept wing.

29

Sweepback of a wing positively influences:

1. static longitudinal stability.
2. static lateral stability.
3. dynamic longitudinal stability.

The combination that regroups all of the correct statements is:

☒ 2.

☐ 1, 2, 3.

☐ 1.

☐ 3.

30

Which aeroplane behaviour will be corrected by a yaw damper?

- ☐ Flutter.
- ☐ Buffeting.
- ☒ Dutch roll.
- ☐ Spiral dive.

31

An example of differential aileron deflection during initiation of left turn is:

- ☐ Left aileron: 2° down.
Right aileron: 5° up.
- ☒ Left aileron: 5° up.
Right aileron: 2° down.
- ☐ Left aileron: 5° down.
Right aileron: 2° up.
- ☐ Left aileron: 2° up.
Right aileron: 5° down.

32

Comparing the differences between a horizontal trimmable stabiliser and an elevator trim tab, which statement is correct?

- I. A stabiliser trim is less suitable for jet transport aeroplanes because of their large speed range.
II. A trim tab is a more powerful means of trimming.

☒ I is incorrect, II is incorrect.

☐ I is correct, II is correct.

☐ I is incorrect, II is correct.

☐ I is correct, II is incorrect.

33

Propeller slip is the difference between the

- ☐ geometric pitch and blade angle of the propeller.
- ☐ plane of rotation of the propeller and forward velocity of the aircraft.
- ☒ geometric pitch and the effective pitch of the propeller.

34

The angle of attack of a fixed pitch propeller blade increases when:

- ☒ RPM increases and forward velocity decreases.
- ☐ forward velocity increases and RPM decreases.
- ☐ velocity and RPM decrease.
- ☐ velocity and RPM increase.

35

Which statement is correct when comparing a fixed pitch propeller with a constant speed propeller?

- I. A constant speed propeller reduces fuel consumption over a range of cruise speeds.
II. A coarse fixed pitch propeller is more efficient during take-off.

- ☐ I is incorrect, II is incorrect.
- ☐ I is correct, II is correct.
- ☒ I is correct, II is incorrect
- ☐ I is incorrect, II is correct.

36

Given:
Aeroplane mass: 50 000kg.
Lift/Drag ratio: 10.
Thrust per engine: 30 000N.
Assumed g: 10m/s².

For a straight, steady, wings level climb of a three-engine aeroplane, the all-engines climb gradient is:

☒ 8.0%.

☐ 2.9%.

☐ 8.5%.

☐ 9.7%.

37

Given:
Aeroplane mass: 50 000kg.
Lift/Drag ratio: 12.
Thrust per engine: 28 000N.
Assumed g: 10m/s².

For a straight, steady, wings level climb of a three-engine aeroplane, the one-engine inoperative climb gradient is:

☐ 8.0%.

☐ 9.7%.

☒ 2.9%.

☐ 8.5%.

38

An aeroplane climbs to cruising level with a constant pitch attitude and maximum climb thrust, (assume no supercharger).
How do the following variables change during the climb? (Gamma = flight path angle)

- ☐ Gamma decreases, angle of attack increases, IAS remains constant.
- ☐ Gamma remains constant, angle of attack remains constant, IAS decreases.
- ☒ Gamma decreases, angle of attack increases, IAS decreases.
- ☐ Gamma decreases, angle of attack remains constant, IAS decreases.

39

Which of the following increases the maximum duration of a glide?

- ☐ A headwind.
- ☐ An increase in mass.
- ☒ A decrease in mass.
- ☐ A tailwind.

Candidate: Denis Mbuya 2
Date of birth: 1986-11-10

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40

(Please view annex 081-0007 issue date October 2005)
The point in the diagram giving the lowest speed in unaccelerated flight is:

☐ point 3.

☒ point 4.

☐ point 2.

☐ point 1.

See Attachments:

1. 081-0007.gif

Annex

(Attachment
No.2)

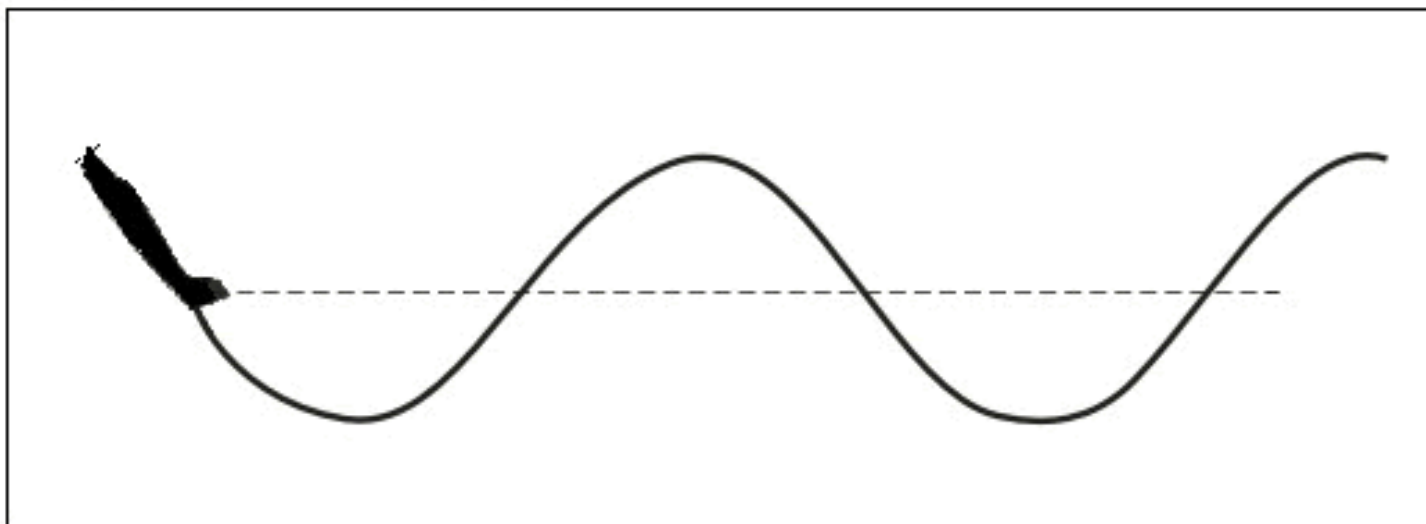
Attachment No. 1

Questions 26

Annex

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081-0023 issue date June 2006



Attachment No. 2

Questions 40

Annex

081-0007.gif

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